

7V0007 MSc Project
NPC ToR Coversheet

Department of Computing and Mathematics Computing and Digital Technology Postgraduate Programmes Terms of Reference Coversheet

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Project title:	Breast Cancer Ultrasound Classification Using Machine Learning
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Signature and date student:	Akerele David Damilola 06/06/2026
Signature and date external collaborator (if involved):	

This sheet should be attached to the front of the completed ToR and uploaded with it to Moodle.

Manchester Metropolitan University

MSc Artificial Intelligence

2025 / 2026

Project Terms of Reference

Breast Cancer Ultrasound Classification Using Machine Learning

Aims & Objectives

Breast cancer is one of the most prevalent and life-threatening cancers worldwide. Early and accurate diagnosis is critical to improving patient survival rates, and medical imaging — particularly ultrasound — is a widely used, non-invasive diagnostic tool. However, the manual interpretation of ultrasound images is time-consuming and subject to inter-observer variability. This project aims to design, develop, and evaluate machine learning models capable of automatically classifying breast cancer ultrasound images into benign, malignant, and normal categories.

The overarching aim is to contribute to the development of an AI-assisted diagnostic tool that can support clinicians in making faster, more consistent, and more accurate breast cancer assessments from ultrasound imagery.

To achieve this aim, the following objectives have been identified:

- Terms of Reference — Define the project scope, aims, objectives, evaluation plan, and activity schedule.
- Literature Review — Conduct a comprehensive survey of existing breast cancer detection methods, machine learning and deep learning architectures, and publicly available ultrasound datasets.
- Data Acquisition & Pre-processing — Identify, obtain, and prepare a suitable labelled ultrasound image dataset (e.g. BUSI dataset), including image normalisation, augmentation, and train/validation/test splitting.
- Model Design & Development — Implement and compare multiple classification approaches, including traditional machine learning classifiers (e.g. SVM, Random Forest) applied to hand-crafted features, and convolutional neural network (CNN) architectures (e.g. ResNet, VGG, EfficientNet) applied to raw image data.
- Evaluation & Testing — Assess model performance using standard metrics: accuracy, precision, recall, F1-score, and AUC-ROC; compare approaches and identify strengths and limitations.
- Final Report — Document all stages of the project, present findings, and critically reflect on outcomes and future directions.

The central research hypothesis guiding the project is:

"Machine learning models, and in particular deep convolutional neural networks, can achieve clinically meaningful classification accuracy on breast cancer ultrasound images, outperforming traditional feature-based approaches."

Learning Outcomes

In accordance with the MSc Artificial Intelligence degree learning outcomes (as specified in the 6G7V0007 Project Handbook, Appendix D), the following outcomes will be addressed and demonstrated through this project:

- Choose, customise, and combine suitable core techniques for solving a given AI problem — the project will evaluate and combine CNN architectures and classical ML pipelines to address the image classification task.
 - Making descriptive and inferential statements about data via appropriate processing techniques and models, accounting for aspects such as uncertainty, data quality, and model fitness — the project will involve statistical analysis of model outputs, uncertainty estimation, and performance benchmarking.
 - Process and analyse, effectively and efficiently, data of varying scales and from heterogeneous sources, formats, and systems, using a range of suitable languages, tools, and environments — the project will involve handling raw medical image data, applying augmentation pipelines, and working across Python, TensorFlow/Keras, and scikit-learn.
 - Build AI solutions with good software practices such as in code reuse, separation of concerns, modularity, testing, and documentation — all code will be structured, version-controlled (GitHub), and accompanied by clear documentation.
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Project Description

Breast cancer ultrasound imaging produces greyscale images of breast tissue; lesions appear as hypoechoic regions and must be classified by their morphological characteristics. Manual classification is challenging and error-prone, making it a strong candidate for AI-assisted analysis.

This project will primarily utilise the Breast Ultrasound Images (BUSI) dataset, a publicly available dataset containing 780 images across three classes: normal, benign, and malignant. The dataset includes segmentation masks alongside images, enabling optional use of region-of-interest (ROI) extraction as a pre-processing step.

The project workflow will proceed through the following stages:

- Data pre-processing: image resizing, pixel normalisation, histogram equalisation, and data augmentation (flipping, rotation, zoom) to address class imbalance and improve model generalisation.
- Feature extraction (for classical ML): texture features using Grey-Level Co-occurrence Matrix (GLCM), shape features, and histogram of oriented gradients (HOG).
- Classical ML pipeline: SVM, Random Forest, and k-NN classifiers trained on extracted features; hyperparameter tuning via cross-validation.
- Deep learning pipeline: transfer learning with pre-trained CNN backbones (ResNet50, VGG16, EfficientNetB0) fine-tuned on the BUSI dataset; custom CNN architecture as a baseline.
- Comparative evaluation: systematic comparison of all approaches using held-out test data; analysis of confusion matrices, ROC curves, and learning curves.

Legal, social, ethical and professional considerations: The BUSI dataset is publicly available for academic research and does not contain personally identifiable information. No human participants will be recruited for this project. The project will comply fully with Manchester Metropolitan University's Academic Ethical Framework. Ethical approval will be obtained via the EthOS system prior to data collection. Findings will be reported honestly and limitations clearly acknowledged, in line with professional computing ethics standards.

References

The following initial references will be used to begin the literature survey:

- Al-Dhabyani, W., Gomaa, M., Khaled, H. and Fahmy, A. (2020) 'Dataset of breast ultrasound images'. Data in Brief, 28, p.104863. [BUSI dataset]
- Shen, L., Margolies, L.R., Rothstein, J.H., Fluder, E., McBride, R. and Sieh, W. (2019) 'Deep learning to improve breast cancer detection on screening mammography'. Scientific Reports, 9(1), p.12495.
- Litjens, G., Kooi, T., Bejnordi, B.E., Setio, A.A.A., Ciompi, F., Ghafoorian, M., van der Laak, J.A., van Ginneken, B. and Sánchez, C.I. (2017) 'A survey on deep learning in medical image analysis'. Medical Image Analysis, 42, pp.60–88.
- He, K., Zhang, X., Ren, S. and Sun, J. (2016) 'Deep residual learning for image recognition'. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp.770–778.
- Tan, M. and Le, Q.V. (2019) 'EfficientNet: Rethinking model scaling for convolutional neural networks'. Proceedings of the 36th International Conference on Machine Learning (ICML), PMLR 97, pp.6105–6114.
- Simonyan, K. and Zisserman, A. (2014) 'Very deep convolutional networks for large-scale image recognition'. arXiv preprint arXiv:1409.1556.
- Cheng, J.Z., Ni, D., Chou, Y.H., Qin, J., Tiu, C.M., Chang, Y.C., Huang, C.S., Shen, D. and Chen, C.M. (2016) 'Computer-aided diagnosis with deep learning architecture: applications to breast lesions in US images and pulmonary nodules in CT scans'. Scientific Reports, 6(1), p.24454.

Evaluation Plan

Evaluation will be conducted at two levels: technical and process-oriented.

Technical evaluation will assess the performance of all developed models against a held-out test set (20% of available data). Metrics will include:

- Overall classification accuracy
- Per-class precision, recall, and F1-score
- Area under the ROC curve (AUC-ROC) for multi-class discrimination
- Confusion matrix analysis to identify misclassification patterns

Model performance will be compared against published benchmarks on the BUSI dataset and against each other to determine whether deep learning approaches offer meaningful advantages over classical ML methods. Ablation studies will be conducted to assess the contribution of individual components (e.g. data augmentation, transfer learning, fine-tuning).

Process evaluation will compare actual progress against the activity schedule, reflecting on deviations and how they were managed. The project will also be self-evaluated against the initial research hypothesis and degree learning outcomes, and suggestions for future work will be made in the final report.

Activity Schedule

The project is planned across the period May 2026 to September 2026. The following table and Gantt chart summarise the key tasks, start and end dates, and durations:

Task	Start Date	End Date	Duration (Days)
Terms of Reference & Ethics	01/05/2026	06/06/2026	36

Literature Review	01/05/2026	15/07/2026	75
Data Acquisition & Pre-processing	07/06/2026	05/07/2026	28
Model Design & Development	06/07/2026	09/08/2026	34
Evaluation & Testing	10/08/2026	24/08/2026	14
Report Writing	20/08/2026	11/09/2026	22

Note: Literature review and data pre-processing tasks overlap intentionally to allow iterative refinement of the approach as relevant papers are identified. Report writing begins before evaluation is complete to allow early drafting of background and methodology chapters.

Signing and Dating

Submission of this document to Moodle by the student counts as the student signature. Acceptance mark on Moodle by the supervisor counts as the supervisor signature, in accordance with the 6G7V0007 Project Handbook.

Student: Akerele David Damilola Date: 06/06/2026

Supervisor: Dr Moi Hoon Yap Date: _____